BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024
SEMESTER –V

Course Code: BCA50113
Course Name: Design and Analysis of Algorithm
Weekly Contact Hours: 4L
Credits: 4
Total Allotted Hours: 60L

Course Objective: This course aims to equip students with a comprehensive understanding of algorithm development, implementation and complexity analysis. Topics include stages of algorithm development, time and space complexity analysis, solving recurrences, searching and sorting algorithms, algorithm design techniques graph and tree algorithms, and complexity classes.

Pre-requisite(s): Data-structure and basic programming.

Course Outcomes (COs): After completion of the course students will be able to:

CO1: Employ problem-solving techniques, analyze time and space complexity, asymptotic notations, and solve recurrences using various methods.
CO2: Evaluate and compare linear search, binary search, merge sort, quick sort, and heap sort complexities.
CO3: Apply brute force, implement divide and conquer, illustrate greedy and dynamic programming, implement backtracking techniques.
CO4: Implement graph traversal, analyze shortest path, and evaluate spanning tree algorithms, enhancing skills in solving graph and tree-related problems.
CO5: Acquire insight into computational complexity classes, differentiate P, NP, and NP-complete classes.

Module 1: Algorithm Development & Complexity Analysis **[10H]**

Details of different asymptotic notations for time and space complexity, time-space trade-off, relationships among different asymptotic notations, their mathematical significance. Solving recurrences: substitution method, recurrence tree method and masters' theorem. Finding time complexities by writing programs on a computer.

Module 2: Searching and Sorting Algorithm **[10H]**

Details of Linear Search and Binary search – their comparisons, Interpolation search and its complexity. Merge sort, quick sort and heap sort, their complexity analysis in best case, worst case and average case.

Module 3: Algorithm Design Techniques **[16H]**

Brute force algorithm; divide and conquer strategy - recursive max-min problem, matrix multiplication using divide and conquer and Strassen's algorithm, greedy techniques - fractional knapsack problem, Job sequencing with deadline. Dynamic programming - 0/1 knapsack problem, matrix chain multiplication. Backtracking-N-queens Problem. String matching algorithm – naïve algorithm, KMP algorithm.

Module 4: Graph and Tree Algorithms **[16H]**

Graph traversal algorithms: Depth First Search (DFS) and Breadth First Search (BFS); shortest path algorithms Dijkstra's algorithm, Bellman Ford algorithm, Floyd-Warshall algorithm, disjoined set data structure, spanning tree Prim's and Krushkal algorithm.

Module 5: Complexity Classes **[8H]**

P class, NP class- Boolean satisfiability problem (SAT), Hamiltonian path problem; NP-complete class - Hamiltonian cycle, vertex cover, cook's theorem, basics of approximation algorithms.

Course Outcomes (COs) and Modules Mapping:

Sl. No.	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K2, K4
2.	CO2	M2	K4, K5
3.	CO3	M3	K2,K3
4.	CO4	M4	K4, K5
5.	CO5	M5	K4,K5

Text books:

1. Introduction to Algorithms, Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022), MIT Press.

Reference books:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and S. Rajasekharan, Universities Press, 2nd edition.
2. The Design and Analysis of Computer Algorithms, Aho, A. V., Hopcroft, J. E., & Ullman, J. D. (1974), Addison-Wesley Series, Pearson.